

AL RAFI AHMED

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EDUCATION

Brac University

Jan 2021 – Jan 2025

Bachelor of Science in Computer Science and Engineering

Merul Badda, Dhaka

CGPA: 3.86/4.00

EXPERIENCE

TallyKhata

Mar 2025 – Jun 2025

Product Engineering Intern

Banani, Dhaka

- Automated PostgreSQL data retrieval and insertion with Python ETL scripts, reducing manual handoffs and speeding up recurring workflows by 30%.
- Built a call analytics pipeline using the Gemini API to extract topics, summaries, discussed issues, and generate actionable insights. Implemented a rate limiter to keep requests within Gemini API quotas while maintaining throughput and stability.
- Investigated user-reported issues on the database; wrote targeted SQL to investigate transaction anomalies, resolved disputes, and supported other daily operational processes.

Brac University

Oct 2023 – Jan 2025

Undergraduate Teaching Assistant

Merul Badda, Dhaka

- Supported instruction and courses taught: Data Structure and Complex Numbers & Laplace Transformations.
- Graded assignments with rubric-based feedback and maintained an accurate mark sheet.
- Provided 15+ hours/week consultation to help the students understand the lab tasks, breakdowns, and code correctness.

PROJECTS

Multilingual RAG System 🐙 | *FastAPI, LangChain, ChromaDB, Sentence Transformers, PyMuPDF, Google Gemini*

- Architected scalable RAG pipeline processing 100+ page documents for Bengali-optimized text segmentation algorithm improving retrieval relevance by 40%.
- Designed responsive React UI with real-time backend status monitoring and document visualization. Implemented persistent vector storage with ChromaDB supporting 10K+ document chunks

AirBNB Rent, a buy/sell platform 🐙 | *ReactJS, ExpressJS, MongoDB, TailwindCSS, OAUTH*

- Developed a property rental and sales platform where users can list, search, book, and track the status of available rooms/houses.
- Implemented secure authentication with Google OAuth and integrated RESTful CRUD APIs for property listings and booking flows on the backend.

Multilevel Image Classification using CNN 🐙 | *Python, Tensorflow, CNN, Pandas, ResNet50, InceptionNet*

- Designed and implemented a deep learning pipeline for classifying images from an unbalanced dataset of 6,500 low-resolution samples. Applied transfer learning using ResNet50 and InceptionNet, achieving 20% higher accuracy compared to baseline CNN models.

SKILLS & PROFICIENCIES

- **Programming Languages:** Python, Java, JavaScript, C++
- **AI & ML Tools:** PyTorch, TensorFlow, Keras, Scikit-learn, Plotly
- **Web Development:** FastAPI, ReactJS, ExpressJS, NodeJS, HTML5, Tailwind CSS, JWT, OAUTH
- **Databases:** MySQL, MongoDB, PostgreSQL, ChromaDB
- **Other Expertise:** Fluent in English and Bengali both writing & speaking

ACHIEVEMENTS AND CERTIFICATIONS

- Placed on VC's list six times and Dean's list twice.
- Receive Merit-Based Scholarship for academic excellence.
- Passed the 'Information Technology Engineers Examination (ITEE) - Level 2 (Fundamental Engineering)' conducted by the BD-ITEC and Bangladesh Computer Council (BCC) in collaboration with the Government of Japan.